

altairsim — CP/M 2.2b on an Altair FDC+

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The **FDC+** is a floppy controller that replaces the 88-DCDD in an Altair. It answers at the same ports, so Altair disk software runs on it, but it can also serve disks from a **drive server** on a PC over a serial cable, and it speaks a 1.5 MB floppy format of its own. Two machines here:

Machine file	What it is
fdcplus-type7.toml	CP/M from an FDC+ Serial Drive Server : no disk in this folder is used. The FDC+ (drive type 7) gets its tracks from the server over a serial cable. Set your serial port in the file first — see below.
fdcplus-type5.toml	CP/M on the FDC+'s 1.5 MB floppy (drive type 5), from CPM22-48K-HDF.dsk . The stock DBL boot PROM boots it, although it cannot read the disk — see below.

To launch one, name the machine file from inside this folder:

```
$ altairsim fdcplus-type5.toml
```

On **Windows** the program is altairsim.exe :

```
> altairsim.exe fdcplus-type5.toml
```

The machine boots itself — you type no **BOOT** command — and comes up at the CP/M prompt.

^E (STOP) takes the keyboard back to the monitor at any point; **RUN** resumes. **^C** belongs to CP/M (it is warm boot) and CP/M gets it.

The files

File	What it is
fdcplus-type7.toml	The serial-drive machine: base = "default" with the 88-DCDD replaced by an FDC+ and a 10 MHz crystal. Read it — it explains why the crystal is not optional.
fdcplus-type5.toml	The 1.5 MB floppy machine: base = "default" with the 88-DCDD replaced by an FDC+ at drive type 5 and the disk in drive 0.
CPM22-48K-HDF.dsk	Mike Douglas's 1.5 MB CP/M 2.2b disk, for a 48K machine: MOVCPM , COPY , PCGET / PCPUT , MBASIC , WM , and games, with 920K free.

There is no undo. Drive 0 is mounted read/write because that is what a real machine is, and CP/M writes to **A:** for anything you create. In a clone `git checkout` puts the image back; in a copy you were handed, nothing does. Copy it first if you are about to test writes in anger.

CP/M on the FDC+'s 1.5 MB floppy

`fdcplus-type5.toml` has the **FDC+** with its drive type switches at 5: a high-density floppy that holds 1.5 MB, one 10,240-byte sector to a track. `CPM22-48K-HDF.dsk` is in drive 0. Run it:

```
$ altairsim fdcplus-type5.toml
```

and you get:

```
48K CP/M 2.2b v1.2
For Altair 1.5Mb Floppy

A>
```

The boot PROM at `FF00` is the same DBL as always, and it cannot read this disk. The FDC+ gives it a sector of its own, from the board's memory, and that sector holds a loader that reads the real disk. The machine file explains it.

Leave the clock at its default. The CP/M for this disk moves each track at a speed that only a 2 MHz processor matches, and at full speed the board keeps 2 MHz time too. A faster `clock_hz` breaks it, as a faster processor would on a real Altair.

CP/M from an FDC+ drive server

`fdcplus-type7.toml` has the **FDC+** in its serial-drive mode. The disk images are on an **FDC+ Serial Drive Server** (or any program that speaks its protocol) at the other end of a serial cable, and the board gets them a track at a time. The simulator and a real FDC+ Altair can use the same images through the same server.

1. Start the server, and mount a bootable 8" Altair CP/M image in its drive 0. No such image is in this folder; `../cadzilla/cpm22b23-56k-drawdemo.dsk` is one (56K CP/M 2.2b v2.3).
2. Open `fdcplus-type7.toml` and set `connect` on `fdc0` to your serial port. The file has examples for macOS (`serial:/dev/cu.usbserial-XXXX`), Linux (`serial:/dev/ttyUSB0`) and Windows (`serial:COM3`).
3. Set `baud` to the rate that the server uses: 9600, 19200, 38400, 57600, 76800, 230400, 403200 or 460800. The two ends must agree.

4. Run it:

```
$ altairsim fdcplus-type7.toml
```

If the port does not open, the error lists the serial ports this computer has. If the port opens but nothing comes after the boot, the drive server is not answering; the machine file turns on the board's `error` debug flag, which reports what the line does wrong.

The file sets `clock_hz = 10000000` and `baud = 230400`. The crystal must not be full speed: the CP/M BIOS gives up on a sector after a count that takes 0.28 seconds at 10 MHz, and a track takes 0.19 seconds at 230400 baud. At full speed the count takes a few milliseconds and no track arrives in time. On a slower line, slow the crystal down too: at 38400 baud a track takes about one second, which needs the real 2 MHz.